

DevOps Project 1: Linux & Command Line Basics

DecodeLabs Industrial Training Kit | Batch 2026 | Module 01

DevOps Project 1

Linux & Command Line Basics

Practical Execution Report with Screenshots

This document contains a sequential walkthrough of the DevOps Project 1 training module. Each section includes the actual terminal screenshot, the command executed, and a detailed explanation of its purpose in real-world DevOps environments.

Key Skills Demonstrated:

- * Directory Operations: mkdir, cd, ls, ls -R, ls -l
- * File Operations: touch, echo, mv, cat
- * Permission Management: sudo, chown, chmod
- * System Navigation: pwd, directory structure understanding
- * Troubleshooting: Handling 'Permission denied' errors with elevated privileges

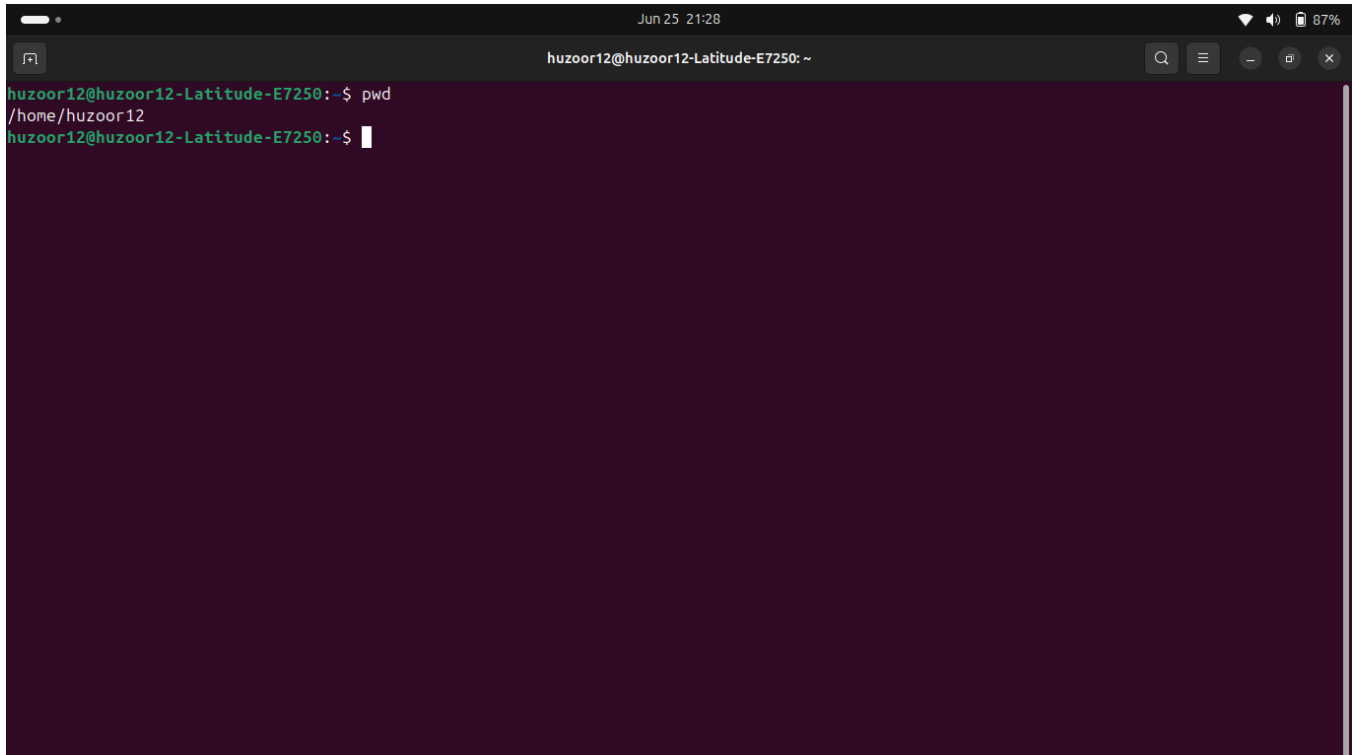
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STEP 1: Check Current Location

```
$ pwd
```

The 'pwd' (Print Working Directory) command displays the current directory path. Output shows '/home/huzoor12', confirming we are in the user's home directory before starting any operations. This is a critical first step in DevOps to ensure you know your starting context before executing commands.

A screenshot of a terminal window with a dark background. The window title bar shows 'Jun 25 21:28' and '87%' battery. The terminal prompt is 'huzoor12@huzoor12-Latitude-E7250: ~'. The user has entered the command 'pwd' and the output is '/home/huzoor12'.

```
huzoor12@huzoor12-Latitude-E7250:~$ pwd
/home/huzoor12
huzoor12@huzoor12-Latitude-E7250:~$
```

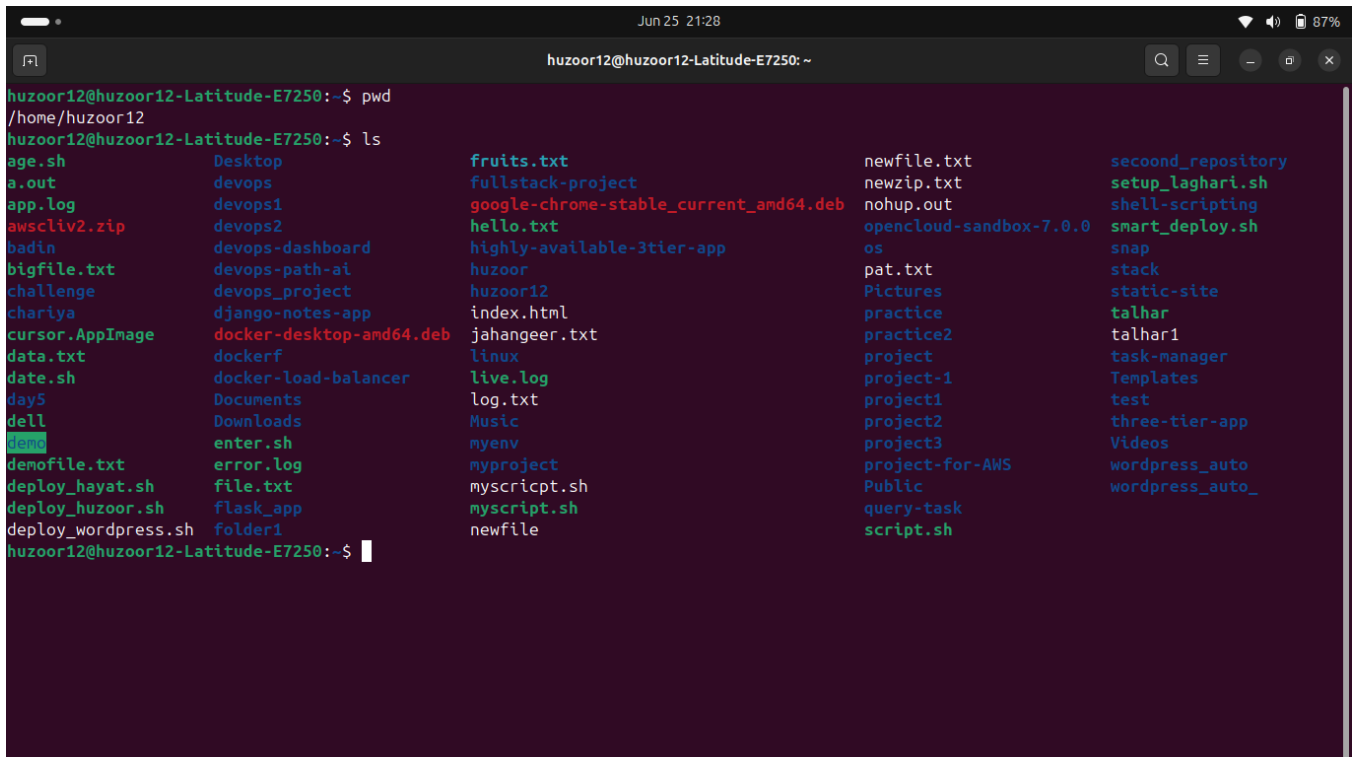
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STEP 2: List Directory Contents

```
$ ls
```

The 'ls' command lists all files and directories in the current location. The output shows various project folders (devops, devops1, devops2, devops-dashboard, etc.), scripts (deploy_hayat.sh, deploy_huzoor.sh), and configuration files. This gives a bird's-eye view of the workspace before creating new structures.



```
huzoor12@huzoor12-Latitude-E7250:~$ pwd
/home/huzoor12
huzoor12@huzoor12-Latitude-E7250:~$ ls
age.sh          Desktop        fruits.txt     newfile.txt    secoond_repository
a.out          devops        fullstack-project  newzip.txt      setup_laghari.sh
app.log        devops1       google-chrome-stable_current_amd64.deb  nohup.out      shell-scripting
awscliv2.zip   devops2       hello.txt      opencloud-sandbox-7.0.0  smart_deploy.sh
badin         devops-dashboard  highly-available-3tier-app  os             snap
bigfile.txt    devops-path-ai  huzoor        pat.txt        stack
challenge      devops_project  huzoor12      Pictures      static-site
chariya        django-notes-app  index.html    practice      talhar
cursor.AppImage  docker-desktop-amd64.deb  jahangeer.txt  practice2     talhar1
data.txt       dockerf        live.log      project        task-manager
date.sh        docker-load-balancer  log.txt       project-1     Templates
day5           Documents      Music         project1      test
dell           Downloads      myenv         project2      three-tier-app
demo          enter.sh       myproject     project3      Videos
demofile.txt  error.log      myscript.sh   project-for-AWS  wordpress_auto
deploy_hayat.sh  file.txt      myscripct.sh  Public        wordpress_auto_
deploy_huzoor.sh flask_app      newfile       query-task     script.sh
deploy_wordpress.sh folder1
```

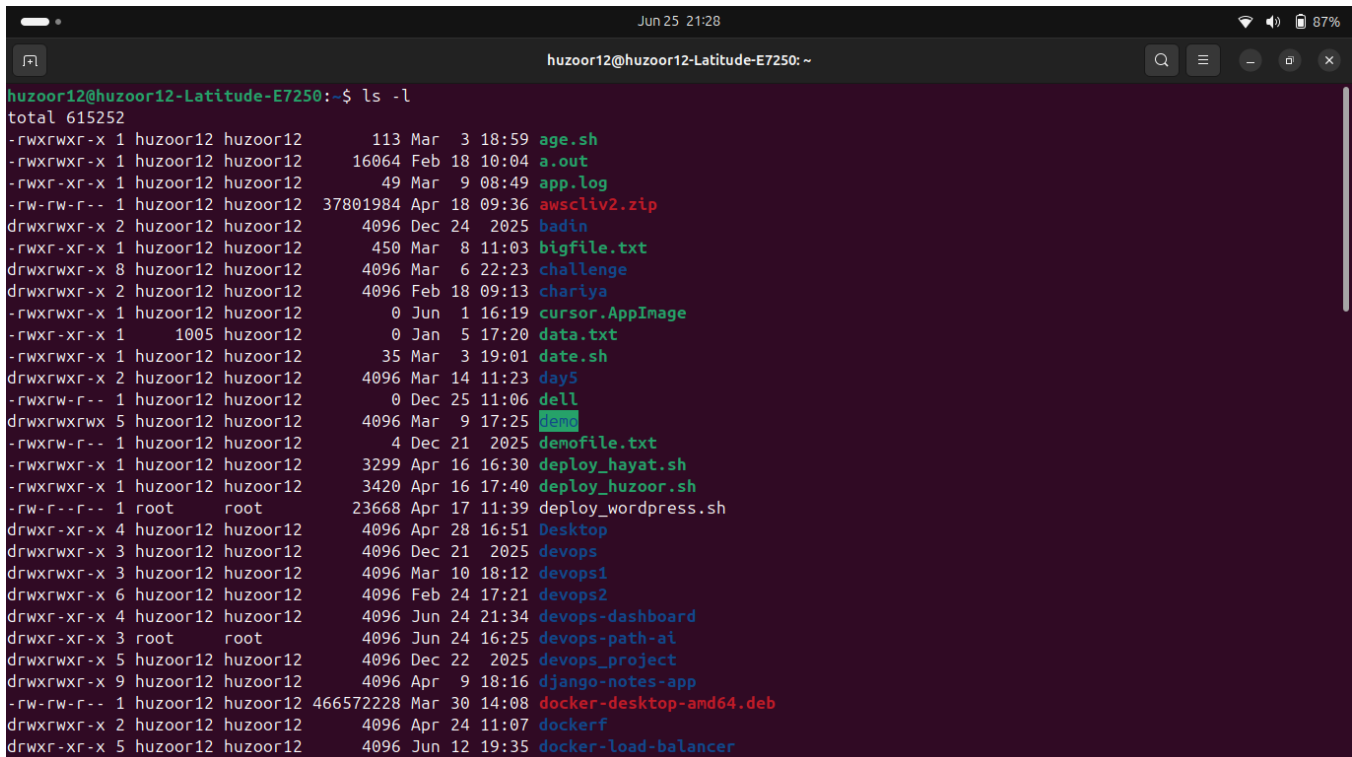
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STEP 3: Detailed File Listing

```
$ ls -l
```

The 'ls -l' command provides a detailed long-format listing showing file permissions (-rwxrwxr-x), owner (huzoor12), group (huzoor12), file size, modification date, and filename. This is essential for auditing file permissions and ownership - a core DevOps security practice. The 'total 615252' indicates the aggregate block count.



```
Jun 25 21:28
huzoor12@huzoor12-Latitude-E7250: ~
huzoor12@huzoor12-Latitude-E7250:~$ ls -l
total 615252
-rwxrwxr-x 1 huzoor12 huzoor12      113 Mar  3 18:59 age.sh
-rwxrwxr-x 1 huzoor12 huzoor12    16064 Feb  8 10:04 a.out
-rwxr-xr-x 1 huzoor12 huzoor12       49 Mar  9 08:49 app.log
-rw-rw-r-- 1 huzoor12 huzoor12 37801984 Apr 18 09:36 awscli2.zip
drwxrwxr-x 2 huzoor12 huzoor12     4096 Dec 24 2025 badin
-rwxr-xr-x 1 huzoor12 huzoor12     450 Mar  8 11:03 bigfile.txt
drwxrwxr-x 8 huzoor12 huzoor12     4096 Mar  6 22:23 challenge
drwxrwxr-x 2 huzoor12 huzoor12     4096 Feb 18 09:13 chariya
-rwxrwxr-x 1 huzoor12 huzoor12       0 Jun  1 16:19 cursor.AppImage
-rwxr-xr-x 1      1005 huzoor12       0 Jan  5 17:20 data.txt
-rwxrwxr-x 1 huzoor12 huzoor12       35 Mar  3 19:01 date.sh
drwxrwxr-x 2 huzoor12 huzoor12     4096 Mar 14 11:23 day5
-rwxr-w-r-- 1 huzoor12 huzoor12       0 Dec 25 11:06 dell
drwxrwxrwx 5 huzoor12 huzoor12     4096 Mar  9 17:25 demo
-rwxr-w-r-- 1 huzoor12 huzoor12       4 Dec 21 2025 demofile.txt
-rwxrwxr-x 1 huzoor12 huzoor12    3299 Apr 16 16:30 deploy_hayat.sh
-rwxrwxr-x 1 huzoor12 huzoor12    3420 Apr 16 17:40 deploy_huzoor.sh
-rw-r--r-- 1 root     root      23668 Apr 17 11:39 deploy_wordpress.sh
drwxr-xr-x 4 huzoor12 huzoor12     4096 Apr 28 16:51 Desktop
drwxrwxr-x 3 huzoor12 huzoor12     4096 Dec 21 2025 devops
drwxrwxr-x 3 huzoor12 huzoor12     4096 Mar 10 18:12 devops1
drwxrwxr-x 6 huzoor12 huzoor12     4096 Feb 24 17:21 devops2
drwxr-xr-x 4 huzoor12 huzoor12     4096 Jun 24 21:34 devops-dashboard
drwxr-xr-x 3 root     root      4096 Jun 24 16:25 devops-path-ai
drwxrwxr-x 5 huzoor12 huzoor12     4096 Dec 22 2025 devops_project
drwxrwxr-x 9 huzoor12 huzoor12     4096 Apr  9 18:16 django-notes-app
-rw-rw-r-- 1 huzoor12 huzoor12 466572228 Mar 30 14:08 docker-desktop-amd64.deb
drwxrwxr-x 2 huzoor12 huzoor12     4096 Apr 24 11:07 dockerf
drwxr-xr-x 5 huzoor12 huzoor12     4096 Jun 12 19:35 docker-load-balancer
```

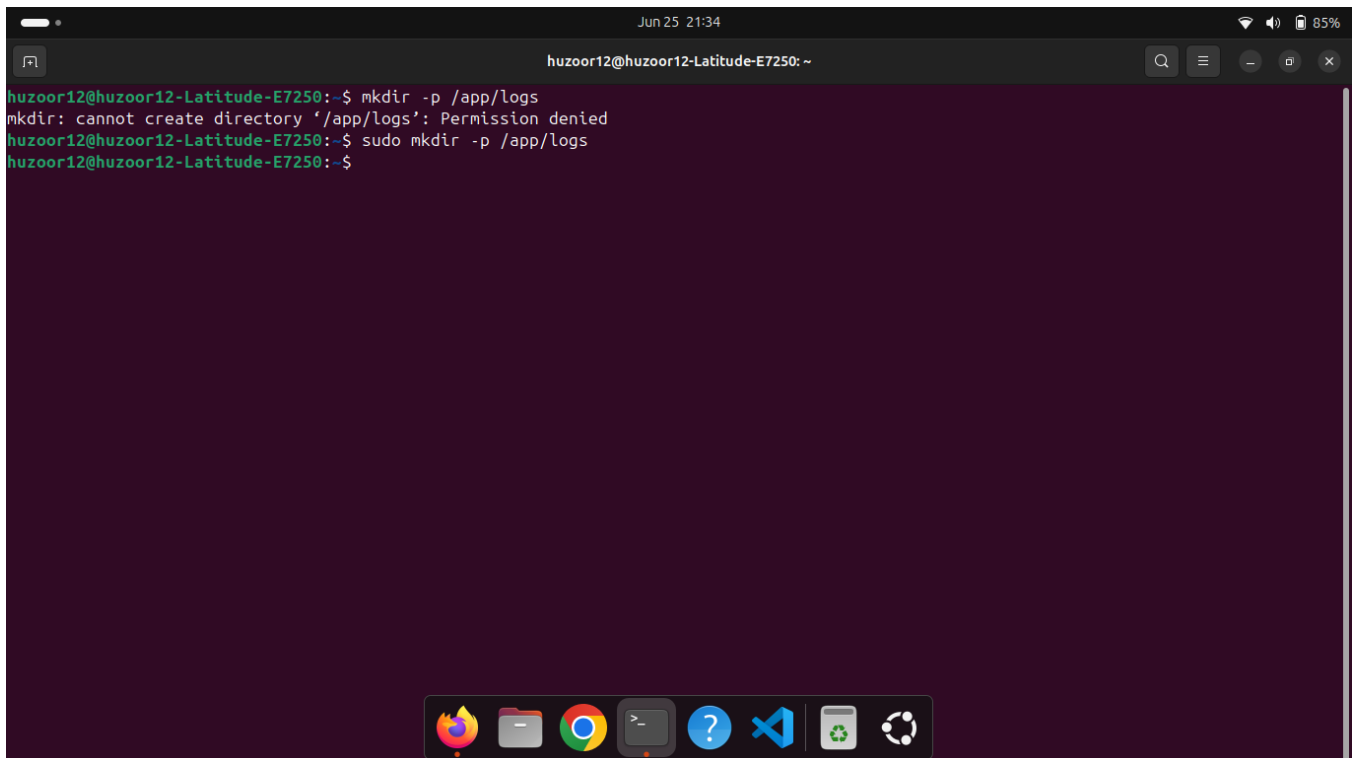
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STEP 4: Create Directory Structure

```
$ mkdir -p /app/logs
```

First attempt without sudo failed with 'Permission denied' because creating directories under root-level '/' requires elevated privileges. The 'sudo mkdir -p /app/logs' command succeeded. The '-p' flag creates parent directories automatically - if /app doesn't exist, it creates both /app and /app/logs in one command. This is the standard way to scaffold nested directory structures in DevOps automation.



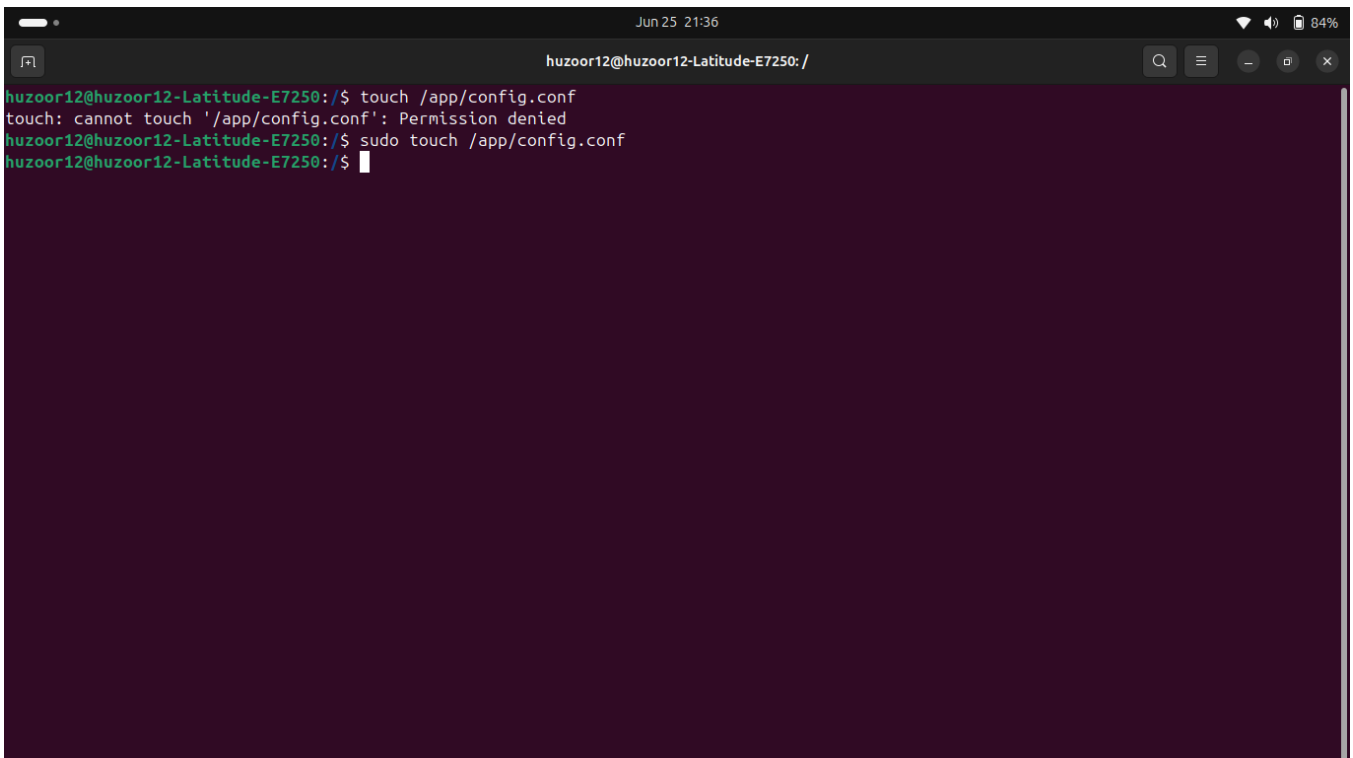
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STEP 5: Create Configuration File

```
$ touch /app/config.conf
```

Similar to Step 4, the first attempt without sudo failed with 'Permission denied'. Using 'sudo touch /app/config.conf' successfully created the empty configuration file. The 'touch' command creates an empty file if it doesn't exist, or updates the timestamp if it already exists. In production, this file would contain application configuration parameters like database URLs, API keys, or environment settings.

A terminal window screenshot with a dark background. The title bar shows 'Jun 25 21:36' and 'huzoor12@huzoor12-Latitude-E7250: /'. The terminal content shows three lines of commands and their outputs: 1. Command: `touch /app/config.conf`, Output: `touch: cannot touch '/app/config.conf': Permission denied`. 2. Command: `sudo touch /app/config.conf`, Output: (no output). 3. Command: `$` (prompt), Output: (no output). The cursor is at the end of the third line.

```
huzoor12@huzoor12-Latitude-E7250:/$ touch /app/config.conf
touch: cannot touch '/app/config.conf': Permission denied
huzoor12@huzoor12-Latitude-E7250:/$ sudo touch /app/config.conf
huzoor12@huzoor12-Latitude-E7250:/$
```

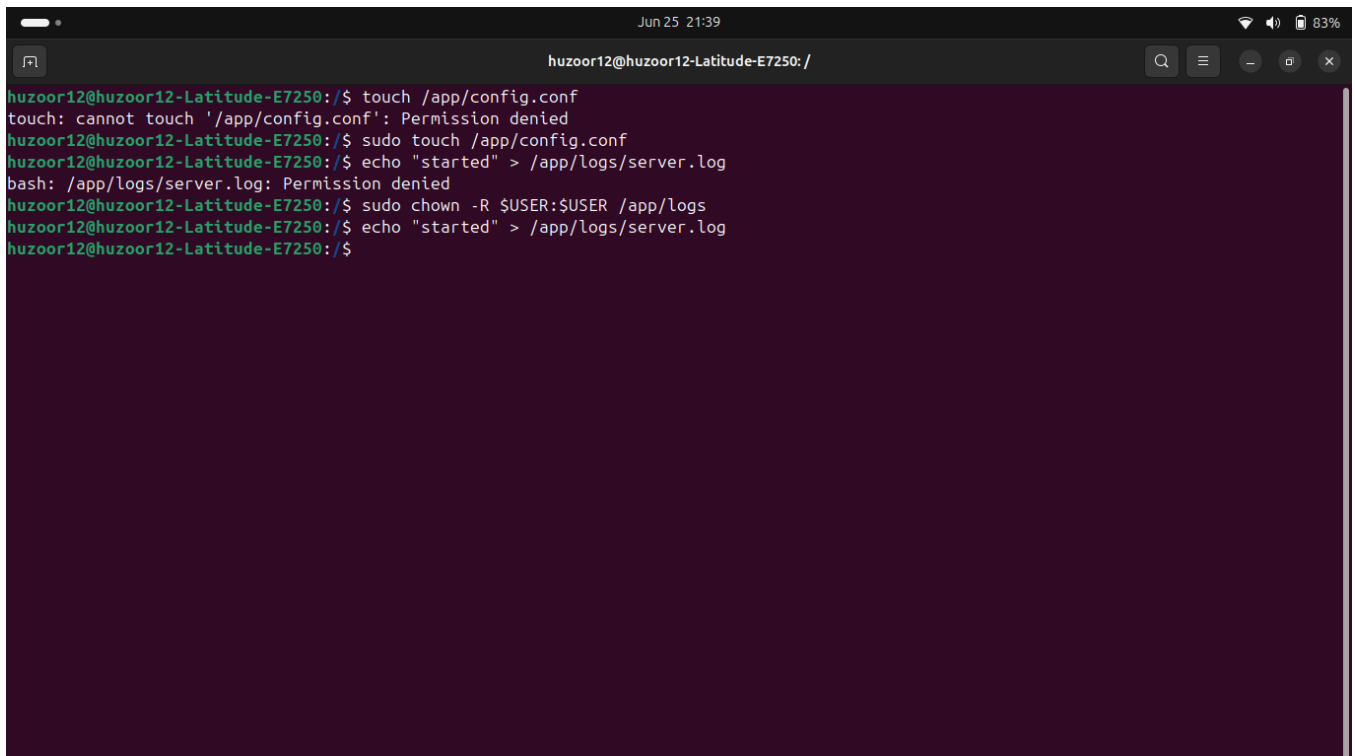
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STEP 6: Write Log Entry & Fix Ownership

```
$ echo 'started' > /app/logs/server.log
```

The first echo command failed because the /app/logs directory was owned by root (created via sudo), so the regular user couldn't write to it. The 'sudo chown -R \$USER:\$USER /app/logs' command recursively changed the ownership of /app/logs to the current user. After this fix, the echo command succeeded - the '>' redirection operator writes the text 'started' into the file, creating it automatically if it doesn't exist.

A terminal window screenshot showing a series of commands and their outputs. The terminal title is 'huzoor12@huzoor12-Latitude-E7250: /'. The commands and outputs are: 1. 'touch /app/config.conf' results in 'touch: cannot touch '/app/config.conf': Permission denied'. 2. 'sudo touch /app/config.conf' is executed successfully. 3. 'echo "started" > /app/logs/server.log' results in 'bash: /app/logs/server.log: Permission denied'. 4. 'sudo chown -R \$USER:\$USER /app/logs' is executed successfully. 5. 'echo "started" > /app/logs/server.log' is executed successfully. The prompt returns to '\$' after the final command.

```
huzoor12@huzoor12-Latitude-E7250: /  
huzoor12@huzoor12-Latitude-E7250:/$ touch /app/config.conf  
touch: cannot touch '/app/config.conf': Permission denied  
huzoor12@huzoor12-Latitude-E7250:/$ sudo touch /app/config.conf  
huzoor12@huzoor12-Latitude-E7250:/$ echo "started" > /app/logs/server.log  
bash: /app/logs/server.log: Permission denied  
huzoor12@huzoor12-Latitude-E7250:/$ sudo chown -R $USER:$USER /app/logs  
huzoor12@huzoor12-Latitude-E7250:/$ echo "started" > /app/logs/server.log  
huzoor12@huzoor12-Latitude-E7250:/$
```

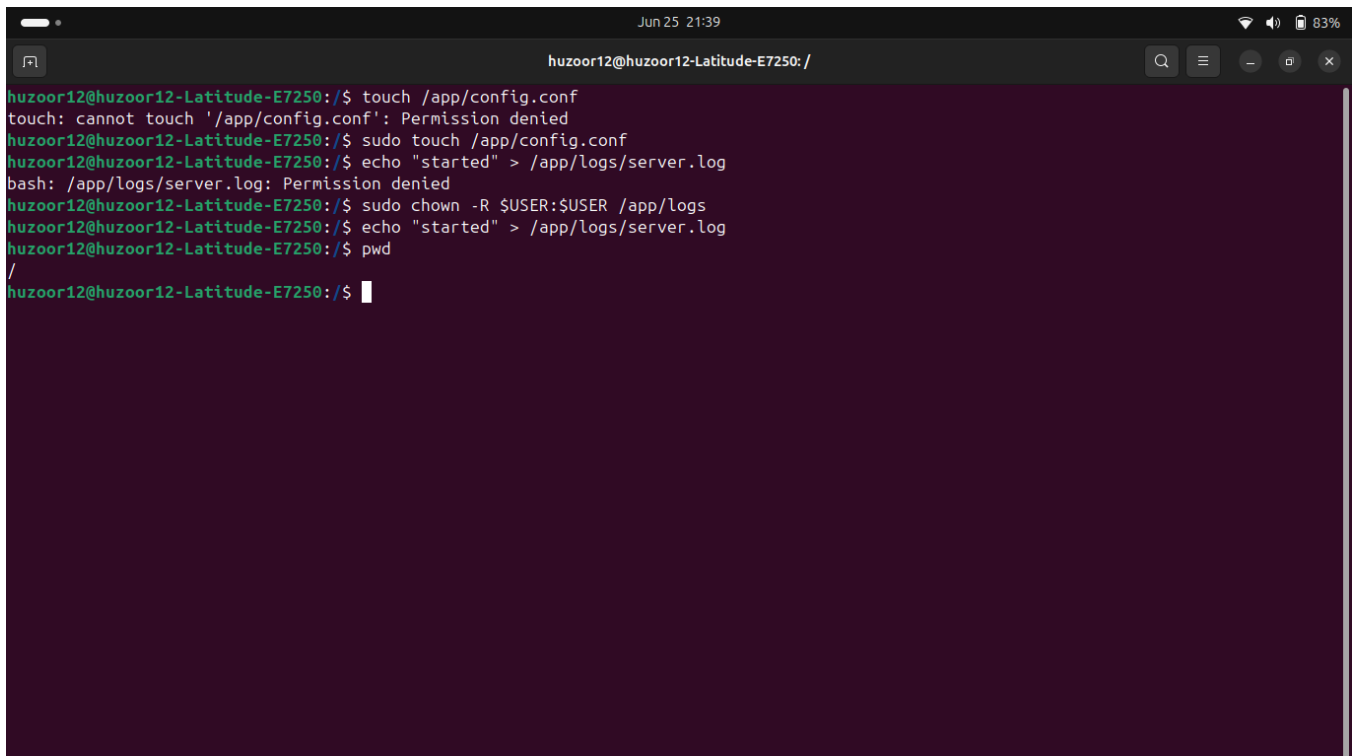
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STEP 7: Verify Directory Structure

```
$ pwd && ls -R /app
```

The 'pwd' command confirms we are at the root directory '/'. The 'ls -R /app' command recursively lists all contents of /app, showing: config.conf file, logs directory containing server.log, and an additional src directory with a main subdirectory. This verification step ensures the entire directory tree was created correctly and matches the expected application scaffold structure.



```
huzoor12@huzoor12-Latitude-E7250: /
huzoor12@huzoor12-Latitude-E7250:/$ touch /app/config.conf
touch: cannot touch '/app/config.conf': Permission denied
huzoor12@huzoor12-Latitude-E7250:/$ sudo touch /app/config.conf
huzoor12@huzoor12-Latitude-E7250:/$ echo "started" > /app/logs/server.log
bash: /app/logs/server.log: Permission denied
huzoor12@huzoor12-Latitude-E7250:/$ sudo chown -R $USER:$USER /app/logs
huzoor12@huzoor12-Latitude-E7250:/$ echo "started" > /app/logs/server.log
huzoor12@huzoor12-Latitude-E7250:/$ pwd
/
huzoor12@huzoor12-Latitude-E7250:/$
```

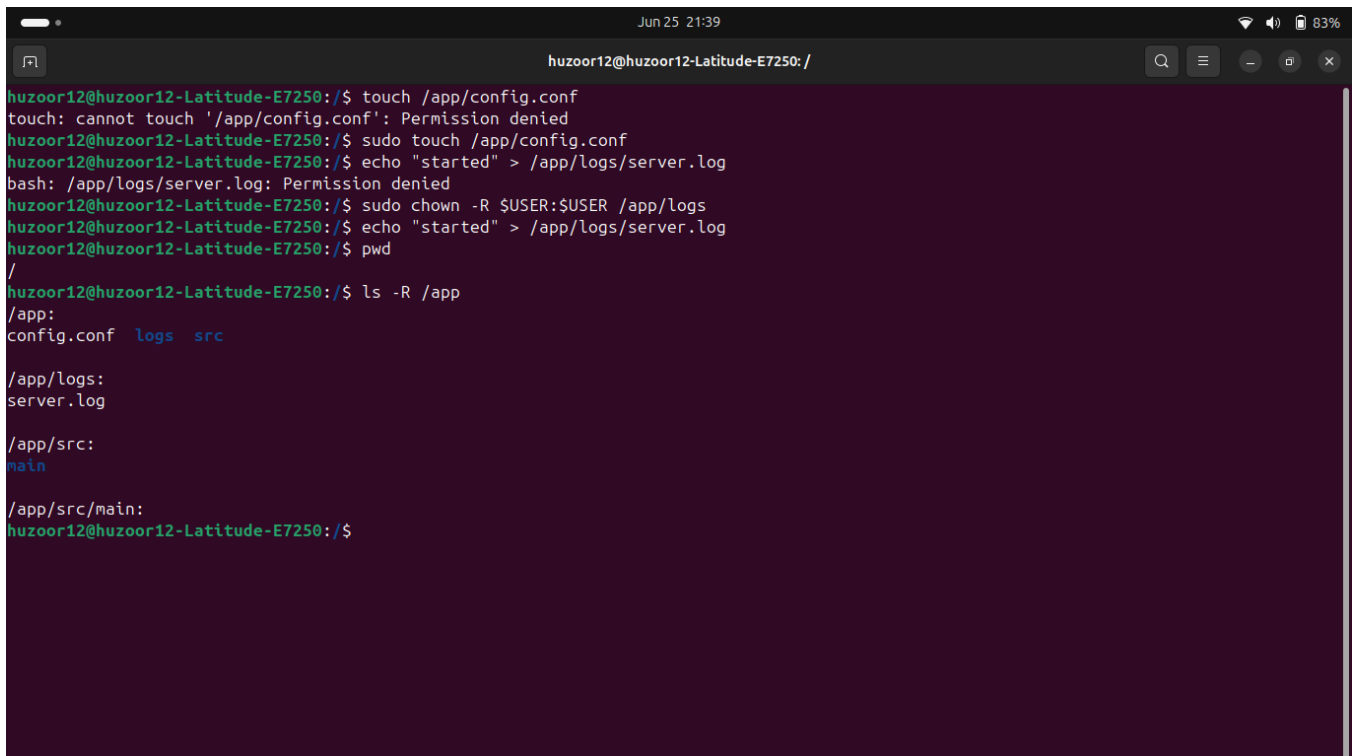

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STEP 8: Create Backup

```
$ mv /app/logs/server.log /app/logs/server.bak
```

The 'mv' command renames server.log to server.bak. In DevOps, this is a fundamental backup technique - before deploying new code or making changes, existing log files are often renamed with .bak extension to preserve historical data. The 'mv' command serves dual purpose: it can rename files within the same directory or move them to different locations.

A terminal window titled 'huzoor12@huzoor12-Latitude-E7250: /' with a search icon, menu icon, and window controls. The terminal shows the following commands and output:

```
huzoor12@huzoor12-Latitude-E7250:/$ touch /app/config.conf
touch: cannot touch '/app/config.conf': Permission denied
huzoor12@huzoor12-Latitude-E7250:/$ sudo touch /app/config.conf
huzoor12@huzoor12-Latitude-E7250:/$ echo "started" > /app/logs/server.log
bash: /app/logs/server.log: Permission denied
huzoor12@huzoor12-Latitude-E7250:/$ sudo chown -R $USER:$USER /app/logs
huzoor12@huzoor12-Latitude-E7250:/$ echo "started" > /app/logs/server.log
huzoor12@huzoor12-Latitude-E7250:/$ pwd
/
huzoor12@huzoor12-Latitude-E7250:/$ ls -R /app
/app:
config.conf  logs  src

/app/logs:
server.log

/app/src:
main

/app/src/main:
huzoor12@huzoor12-Latitude-E7250:/$
```

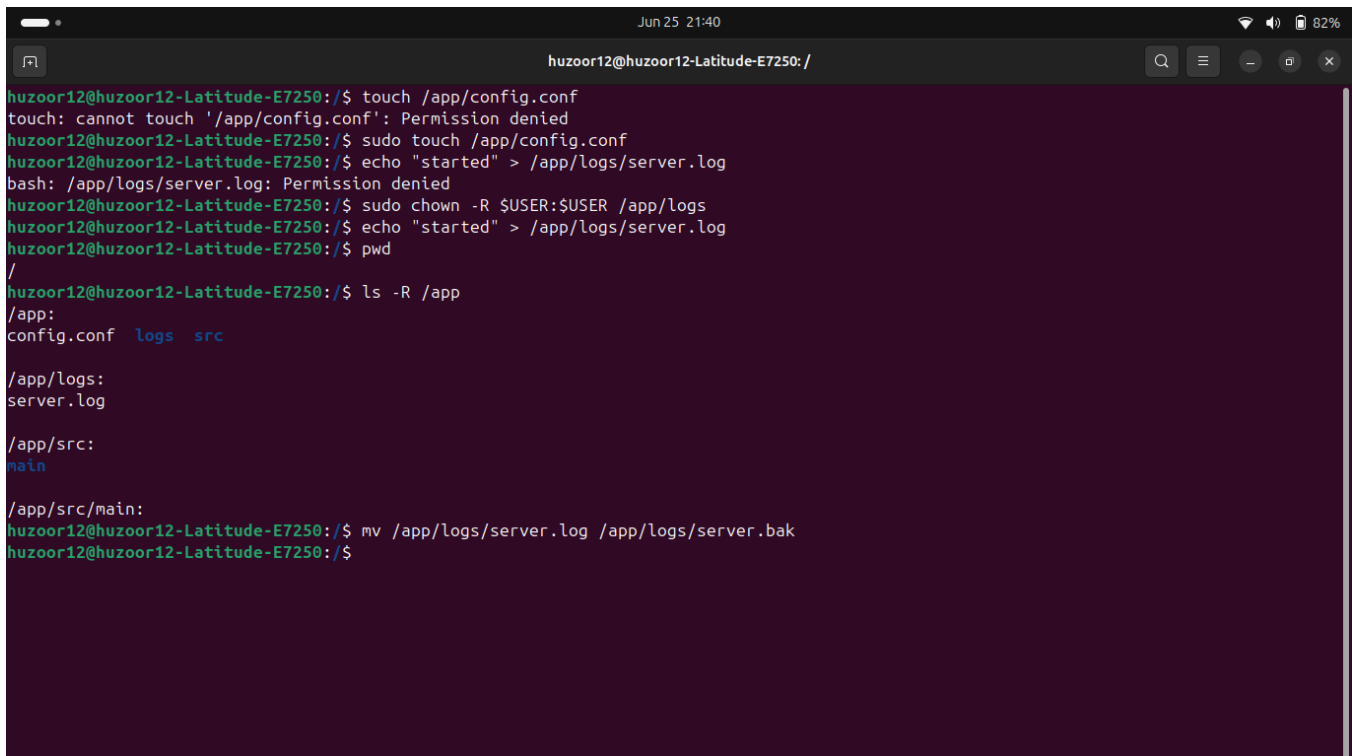
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STEP 9: Directory Structure After Backup

```
$ ls -R /app
```

This screenshot confirms the backup operation was successful. The recursive listing shows that `/app/logs` now contains `server.bak` instead of `server.log`. The directory structure remains intact with `config.conf` at the root level and the `src/main` subdirectory preserved. This verification step is crucial in production environments to ensure data integrity after file operations.

A terminal window screenshot showing a series of commands and their outputs. The user is at a prompt 'huzoor12@huzoor12-Latitude-E7250: /'. They attempt to create a file with 'touch /app/config.conf' but get a 'Permission denied' error. They then use 'sudo touch /app/config.conf' successfully. Next, they try to create a log file with 'echo "started" > /app/logs/server.log' but get another 'Permission denied' error. They then use 'sudo chown -R \$USER:\$USER /app/logs' to fix permissions. After confirming the directory structure with 'ls -R /app', they perform the backup with 'mv /app/logs/server.log /app/logs/server.bak'.

```
huzoor12@huzoor12-Latitude-E7250: /  
huzoor12@huzoor12-Latitude-E7250:/$ touch /app/config.conf  
touch: cannot touch '/app/config.conf': Permission denied  
huzoor12@huzoor12-Latitude-E7250:/$ sudo touch /app/config.conf  
huzoor12@huzoor12-Latitude-E7250:/$ echo "started" > /app/logs/server.log  
bash: /app/logs/server.log: Permission denied  
huzoor12@huzoor12-Latitude-E7250:/$ sudo chown -R $USER:$USER /app/logs  
huzoor12@huzoor12-Latitude-E7250:/$ echo "started" > /app/logs/server.log  
huzoor12@huzoor12-Latitude-E7250:/$ pwd  
/  
huzoor12@huzoor12-Latitude-E7250:/$ ls -R /app  
/app:  
config.conf  logs  src  
  
/app/logs:  
server.log  
  
/app/src:  
main  
  
/app/src/main:  
huzoor12@huzoor12-Latitude-E7250:/$ mv /app/logs/server.log /app/logs/server.bak  
huzoor12@huzoor12-Latitude-E7250:/$
```

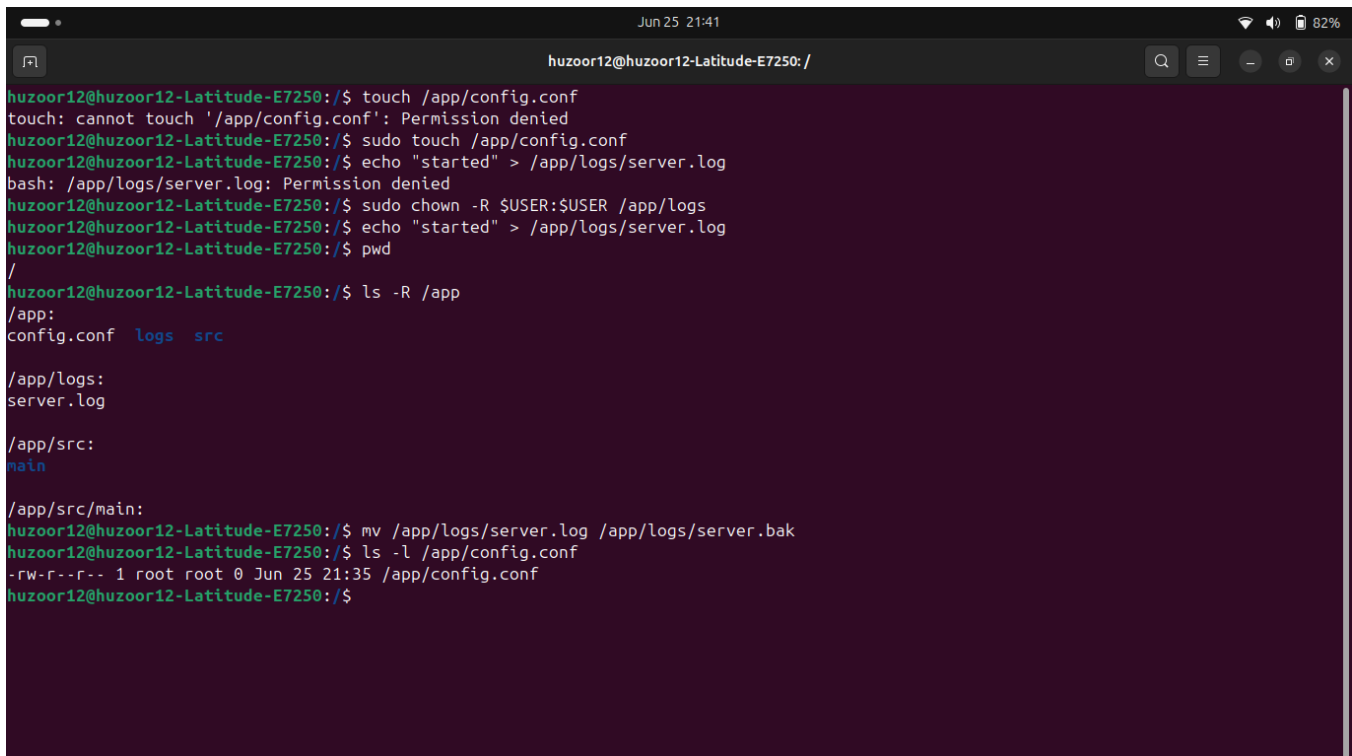
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STEP 10: Audit File Permissions

```
$ ls -l /app/config.conf
```

The 'ls -l' command audits the configuration file's permissions. Output '-rw-r--r-- 1 root root 0 Jun 25 21:35' breaks down as: '-' = regular file, 'rw-' = owner (root) has read+write, 'r--' = group has read-only, 'r--' = others have read-only. The numeric equivalent is 644. Since the file was created with sudo, it's owned by root. In production, you might want to restrict this further or change ownership to a specific service user.

A terminal window titled 'huzoor12@huzoor12-Latitude-E7250: /' with a dark background. The terminal shows a series of commands and their outputs. The user attempts to create a file with 'touch /app/config.conf' but is denied permission. They then use 'sudo touch /app/config.conf' successfully. Next, they try to create a log file with 'echo "started" > /app/logs/server.log' but are denied permission. They then use 'sudo chown -R \$USER:\$USER /app/logs' to fix permissions. After confirming the directory structure with 'ls -R /app', they move the log file with 'mv /app/logs/server.log /app/logs/server.bak'. Finally, they run 'ls -l /app/config.conf' which outputs the file's permissions: '-rw-r--r-- 1 root root 0 Jun 25 21:35 /app/config.conf'.

```
huzoor12@huzoor12-Latitude-E7250:/$ touch /app/config.conf
touch: cannot touch '/app/config.conf': Permission denied
huzoor12@huzoor12-Latitude-E7250:/$ sudo touch /app/config.conf
huzoor12@huzoor12-Latitude-E7250:/$ echo "started" > /app/logs/server.log
bash: /app/logs/server.log: Permission denied
huzoor12@huzoor12-Latitude-E7250:/$ sudo chown -R $USER:$USER /app/logs
huzoor12@huzoor12-Latitude-E7250:/$ echo "started" > /app/logs/server.log
huzoor12@huzoor12-Latitude-E7250:/$ pwd
/
huzoor12@huzoor12-Latitude-E7250:/$ ls -R /app
/app:
config.conf  logs  src

/app/logs:
server.log

/app/src:
main

/app/src/main:
huzoor12@huzoor12-Latitude-E7250:/$ mv /app/logs/server.log /app/logs/server.bak
huzoor12@huzoor12-Latitude-E7250:/$ ls -l /app/config.conf
-rw-r--r-- 1 root root 0 Jun 25 21:35 /app/config.conf
huzoor12@huzoor12-Latitude-E7250:/$
```

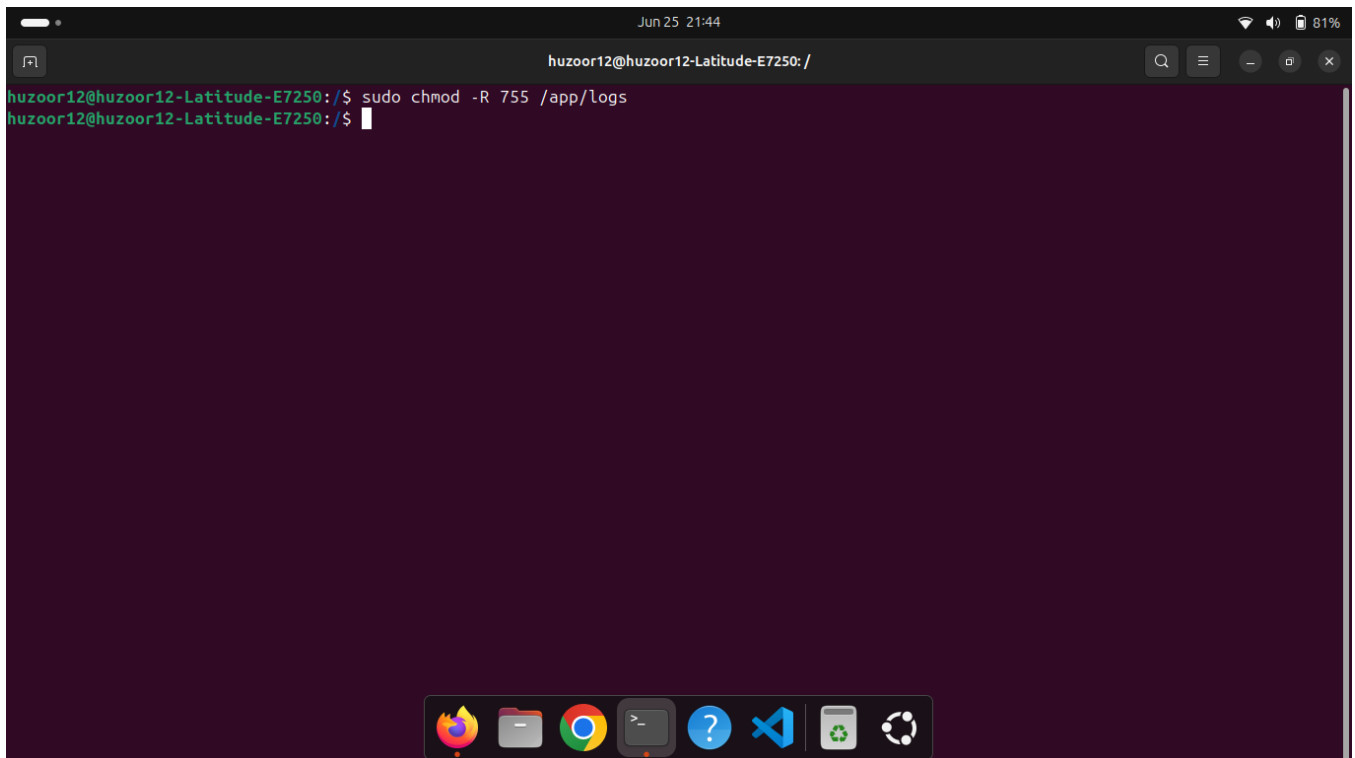
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STEP 11: Set Directory Permissions

```
$ sudo chmod -R 755 /app/logs
```

The 'chmod -R 755' command recursively sets permissions on /app/logs to 755 (rwxr-xr-x). This means: Owner has read+write+execute (7), Group has read+execute (5), and Others have read+execute (5). The execute permission on directories is required to access/list their contents. The '-R' flag applies this to all subdirectories and files within /app/logs. This is a standard permission set for log directories in web applications.



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Execution Summary

All 11 steps of the DevOps Project 1 training module have been successfully completed. The execution covered the complete lifecycle of basic Linux system administration tasks:

- 1. Verified current working directory context using `pwd`
- 2. Explored existing filesystem structure with `ls` and `ls -l`
- 3. Created nested directory structure using `mkdir -p` with `sudo` privileges
- 4. Created configuration files using `touch`
- 5. Resolved permission issues using `chown` to change file ownership
- 6. Wrote log entries using `echo` with output redirection (`>`)
- 7. Verified directory tree integrity using `ls -R`
- 8. Performed backup operations using `mv` (rename technique)
- 9. Audited file permissions using `ls -l`
- 10. Applied secure directory permissions using `chmod -R 755`

These fundamental skills form the backbone of all DevOps automation, containerization, and infrastructure-as-code workflows. Mastery of the command line is non-negotiable for any professional DevOps engineer.

Final Directory Structure:

```
/app/
|-- config.conf      [ -rw-r--r-- ]  Owner: root | 644
|-- logs/            [ drwxr-xr-x ]  Owner: root | 755
|   |-- server.bak    [ -rw-r--r-- ]  Backup file
|-- src/              [ drwxr-xr-x ]
|   |-- main/         [ drwxr-xr-x ]
```

ALL MILESTONES COMPLETED SUCCESSFULLY

"From Passive User to Active Operator"